

ASSIGNMENT 1 : C++ Sample Run

COMP 371: Computer Graphics

Professor : Nagi Basha

Arina Sabzevari (40205861)

Parsa Ghadimi (40203370)

Zineb Alaoui Aziz (40215844)

Gina Cody School of Engineering and Computer Science

Concordia University

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We certify that this submission is the original work of members of the group and
meets the Faculty's Expectations of Originality.

PART 1: Array

```
== Assignment 1 : Part 1 ==  
Enter size: 10  
Making an array of size 10...  
Array created.  
Array is now filled with numbers.  
Array content: [ 0 1 2 3 4 5 6 7 8 9 ]  
Removing the array...  
Memory cleared.  
Done.
```

PART 2: Point and Triangle

- Creation of 3 points :

```
== Assignment 1 : Part 2 ==  
  
Enter coordinates for Vertex 1:  
  x = 2  
  y = 2  
  z = 2  
  
Enter coordinates for Vertex 2:  
  x = 5  
  y = 5  
  z = 5  
  
Enter coordinates for Vertex 3:  
  x = 2  
  y = 9  
  z = 3  
  
Triangle successfully created.
```

- Menu Displayed :

```
=====
      TRIANGLE PROGRAM
=====
1. Show Triangle Coordinates
2. Move a Single Point
3. Move Entire Triangle
4. Calculate Triangle Area
5. Exit
=====
Select an option: █
```

- Choice 1:

```
=====
      TRIANGLE PROGRAM
=====
1. Show Triangle Coordinates
2. Move a Single Point
3. Move Entire Triangle
4. Calculate Triangle Area
5. Exit
=====
Select an option: 1

Current Triangle Coordinates:
Triangle:
Vertex 1: (2, 2, 2)
Vertex 2: (5, 5, 5)
Vertex 3: (2, 9, 3)
```

- Choice 2 :

```

=====
          TRIANGLE PROGRAM
=====
1. Show Triangle Coordinates
2. Move a Single Point
3. Move Entire Triangle
4. Calculate Triangle Area
5. Exit
=====
Select an option: 2

Which point do you want to move? (1, 2, or 3): 1
Enter movement distance: 5
Enter axis to move along (x, y, or z): x
Point moved successfully.

Updated Triangle Coordinates:
Triangle:
Vertex 1: (7, 2, 2)
Vertex 2: (5, 5, 5)
Vertex 3: (2, 9, 3)

```

- Choice 3 :

```

=====
          TRIANGLE PROGRAM
=====
1. Show Triangle Coordinates
2. Move a Single Point
3. Move Entire Triangle
4. Calculate Triangle Area
5. Exit
=====
Select an option: 3

Enter movement distance: 6
Enter axis to move along (x, y, or z): z
Triangle translated successfully.

Updated Triangle Coordinates:
Triangle:
Vertex 1: (7, 2, 8)
Vertex 2: (5, 5, 11)
Vertex 3: (2, 9, 9)

```

- Choice 4 :

```
=====
      TRIANGLE PROGRAM
=====
1. Show Triangle Coordinates
2. Move a Single Point
3. Move Entire Triangle
4. Calculate Triangle Area
5. Exit
=====
Select an option: 4

The area of the triangle is: 11.1131 units2
```

- **Choice 5 :**

```
=====
      TRIANGLE PROGRAM
=====
1. Show Triangle Coordinates
2. Move a Single Point
3. Move Entire Triangle
4. Calculate Triangle Area
5. Exit
=====
Select an option: 5

Exiting program... Thank you.
```